

LA BAIE, QC, Canada

WMO: 713880

Lat: 48.2981N

Lon: 70.9194W

Elev: 498

StdP: 14.43

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.5	-17.8	-30.3	1.0	-21.9	-25.4	1.4	-17.4	26.8	6.2	24.8	4.7	5.1	150	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	84.2	67.4	80.6	66.4	77.3	64.9	70.9	79.4	69.0	76.7	67.1	74.2	9.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	105.5	74.6	66.3	98.8	72.5	64.4	92.4	70.6	35.1	78.8	33.5	77.1	32.0	74.1	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	20.1	17.8	DB	-30.5	91.2	5.8	3.1	-34.7	93.4	-38.1	95.2	-41.3	97.0	-45.5	99.2	(4)
(5)			WB	-30.7	75.0	5.7	2.3	-34.9	76.7	-38.2	78.0	-41.4	79.3	-45.6	80.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	4.9	8.5	19.7	35.3	50.5	59.3	64.2	62.8	54.6	42.1	29.7	14.4	
	DBStd	22.74	13.98	12.79	12.06	8.16	7.87	6.81	5.24	5.44	7.50	7.39	9.21	12.48	
	HDD50	6057	1398	1163	941	443	91	7	0	0	36	265	610	1103	
	HDD65	10238	1863	1583	1406	892	453	197	78	108	321	710	1059	1568	
	CDD50	1430	0	0	0	2	107	285	442	397	175	21	1	0	
	CDD65	135	0	0	0	0	4	25	55	40	10	1	0	0	
	CDH74	1527	0	0	0	2	149	343	534	376	118	5	0	0	
	CDH80	365	0	0	0	0	39	91	131	83	21	0	0	0	
Wind	WSAvg	7.5	8.0	8.3	8.7	8.8	7.9	7.0	6.2	6.1	6.4	7.5	7.8	7.6	
Precipitation	PrecAvg	37.7	2.4	2.1	2.2	2.3	3.1	3.6	4.6	4.0	4.0	3.3	3.0	3.0	
	PrecMax	48.0	4.8	4.4	5.0	4.2	7.1	6.3	12.6	9.6	8.9	5.9	5.8	5.6	
	PrecMin	29.7	0.7	0.5	0.5	0.5	0.6	0.7	1.5	1.4	1.6	1.1	0.9	1.1	
	PrecStd	4.1	0.9	0.9	1.0	1.0	1.3	1.3	1.8	1.6	1.4	1.1	1.0	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	39.3	49.4	65.2	85.0	87.9	88.1	87.2	82.9	71.3	57.4	40.9	
		MCWB	37.0	35.6	41.5	50.7	60.9	68.6	70.1	69.9	67.9	59.6	51.6	38.0	
	2%	DB	35.1	34.5	42.6	57.8	78.0	82.3	83.7	81.9	77.4	64.2	48.9	36.3	
		MCWB	34.0	32.2	36.5	45.4	58.9	65.1	68.6	68.0	66.3	56.2	45.0	34.3	
	5%	DB	30.9	32.1	39.1	52.1	72.0	77.7	79.7	78.0	72.3	57.6	44.6	33.8	
		MCWB	29.4	30.3	34.7	43.0	56.3	63.9	67.5	66.0	63.0	51.6	42.0	32.3	
	10%	DB	25.7	29.0	36.3	47.3	66.3	73.6	76.4	74.6	67.9	53.4	41.3	31.6	
		MCWB	24.2	27.2	33.3	40.3	53.9	61.2	65.7	64.5	61.0	49.1	39.1	30.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	36.8	44.4	53.1	66.6	72.4	74.2	72.5	71.1	61.7	52.2	38.5
			MCDB	38.5	38.1	47.6	62.3	77.6	83.3	83.2	81.9	79.7	69.4	56.8	40.1
2%		WB	33.9	32.8	37.8	46.7	62.1	68.7	71.1	70.0	67.7	56.5	45.8	34.8	
		MCDB	35.0	34.3	40.9	55.0	72.6	77.3	78.7	78.5	74.3	62.8	48.2	36.0	
5%		WB	29.5	30.3	35.6	43.8	57.9	66.0	69.2	68.1	64.4	52.6	42.1	32.7	
		MCDB	30.8	32.1	38.4	50.5	68.5	74.0	76.8	75.1	71.0	56.5	44.1	33.7	
10%		WB	24.2	27.0	33.4	41.1	55.0	63.6	67.2	66.2	61.6	49.4	39.1	30.4	
		MCDB	25.7	28.9	36.0	46.5	64.8	70.8	74.2	72.3	66.8	52.9	41.1	31.6	
Mean Daily Temperature Range		5% DB	MDBR	18.9	20.2	20.0	17.4	22.5	21.6	20.7	21.0	20.3	15.0	12.2	14.7
			MCDBR	19.4	17.4	22.2	26.6	34.3	29.9	27.2	27.2	27.6	23.7	16.7	14.2
	MCWBR		18.5	15.8	17.2	17.1	17.3	15.9	14.4	14.7	16.7	16.2	13.9	13.6	
	5% WB	MCDBR	19.2	16.7	17.7	23.1	28.0	23.8	23.2	22.3	24.4	21.5	15.5	14.1	
		MCWBR	18.3	15.4	14.3	16.1	15.8	14.4	13.8	13.2	15.7	15.6	13.3	13.7	
Clear-Sky Solar Irradiance	taub	0.265	0.284	0.316	0.382	0.363	0.385	0.405	0.385	0.372	0.330	0.313	0.267		
	taud	2.273	2.225	2.198	2.189	2.396	2.365	2.365	2.320	2.380	2.403	2.505	2.395	2.322	
	Ebn at Noon	256	276	284	274	285	278	270	271	264	259	235	240		
	Edh at Noon	23	31	38	43	36	37	39	35	31	24	21	20		
All-Sky Solar Radiation	RadAvg	417	721	1114	1413	1632	1742	1649	1405	1078	619	339	276		
	RadStd	51	94	92	111	158	117	163	93	81	62	35	29		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (27 neighbors)		(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)	(u)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page